

Amadey - APT-C-36 Lab

端点检测与响应（EDR）系统发出的非工作时间警报会标记 Windows 工作站上的可疑活动。被标记的恶意软件与 Amadey 木马窃取者相符。你的任务是分析呈现的内存转储，并为恶意软件所采取的操作创建详细报告。

1. 在内存倾印分析中，确定恶意活动的根源对于理解入侵的范围至关重要。触发这种恶意行为的父进程叫什么名字？

```
vol.py -f Windows\7\ x64-Snapshot4.vmem windows.pslist
```

能看到有个 lssass.exe ,pid2748 的 ,这是本地安全认证子系统服务的伪装 ,真正应为 lsass.exe 。

```
ubuntu@ip-172-31-22-196:~/Desktop/Start here/Artifacts$ vol.py -f Windows\7\ x64-Snapshot4.vmem windows.pslist
```

PID	PPID	ImageFileName	PDB scanning finished	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
2748	2524	lsass.exe	0x16a7750	7	254	1	True	2023-08-09 21:33:04.000000	N/A	Disabled	
3864	2748	rundll32.exe	0x1955b30	1	64	1	True	2023-08-09 21:33:56.000000	N/A	Disabled	
2508	500	wmpnetwk.exe	0x1a2cb30	10	288	0	False	2023-08-09 21:32:17.000000	N/A	Disabled	
2476	1604	cmd.exe	0x36721b0	0	0	False	2023-08-09 21:50:06.000000	2023-08-09 21:50:07.000000	Disabled		Disabled
2124	924	taskeng.exe	0x4bf28e0	5	85	1	False	2023-08-09 21:34:00.000000	N/A	Disabled	
1604	500	vmtoolsd.exe	0x58eb060	9	305	0	False	2023-08-09 21:32:11.000000	N/A	Disabled	

2. 一旦识别出恶意进程，它在设备上的确切位置就能揭示更多关于其性质和来源的信息。这个进程位于工作站的哪个位置？

```
vol.py -f Windows\7\ x64-Snapshot4.vmem windows.cmdline --pid=2748
```

```
ubuntu@ip-172-31-22-196:~/Desktop/Start here/Artifacts$ vol.py -f Windows\7\ x64-Snapshot4.vmem windows.cmdline --pid=2748
```

PID	Process Args
2748	lsass.exe "C:\Users\0XSH3R-1\AppData\Local\Temp\925e7e99c5\lsass.exe"

3. 持续的外部通信表明该恶意软件试图连接 C2C 服务器。您能否确定该进程与之交互的命令与控制（C2C）服务器的 IP 地址？

```
vol.py -f Windows\7\ x64-Snapshot4.vmem windows.netscan
```

0x1e6db500	UDPv4	0.0.0.0	53408	*	0	368	svchost.exe	2023-08-09 21:32:18.000000
0x1e928010	TCPv4	0.0.0.0	49155	0.0.0.0	0	LISTENING	500	services.exe -
0x1e928010	TCPv6	::	49155	::	0	LISTENING	500	services.exe -
0x1e94dcf0	TCPv4	192.168.195.136	49168	41.75.84.12	80	CLOSED	2748	lsass.exe N/A
0x1e9846b0	UDPv4	192.168.195.136	68	*	0	768	svchost.exe	2023-08-09 21:50:07.000000
0x1e988840	UDPv4	0.0.0.0	5355	*	0	920	svchost.exe	2023-08-09 21:50:07.000000
0x1e988840	UDPv6	::	5355	*	0	920	svchost.exe	2023-08-09 21:50:07.000000
0x1ef43830	UDPv6	::1	53403	*	0	2620	svchost.exe	2023-08-09 21:32:17.000000
0x1f1dd690	TCPv4	0.0.0.0	5357	0.0.0.0	0	LISTENING	4	System -
0x1f1dd690	TCPv6	::	5357	::	0	LISTENING	4	System -

4. 根据恶意软件与 C2C 服务器的连接，该恶意软件很可能正在获取其他工具或模块。它试图将多少个不同的文件引入到受感染的工作站？

vol.py -f Windows\ 7\ x64-Snapshot4.vmem windows.memmap.Memmap --pid 2748 --dump
 windows.memmap.Memmap --pid 2748 --dump :将 pid2748 的进程所使用的物理内存数据全部
 转储出来

```
0xfa80107ff000 0x1cde8000 0x1000 0xa5d4000 pid.2748.dmp
0xfa8010908000 0x1cce0000 0x1000 0xa5d5000 pid.2748.dmp
0xfa80109b3000 0x1d04b000 0x1000 0xa5d6000 pid.2748.dmp
0xfa80109fa000 0x1c155000 0x1000 0xa5d7000 pid.2748.dmp
0xfa80109fb000 0x1ce56000 0x1000 0xa5d8000 pid.2748.dmp
0xfa80109fc000 0x1ce17000 0x1000 0xa5d9000 pid.2748.dmp
0xfa8010a89000 0x1cce2000 0x1000 0xa5da000 pid.2748.dmp
0xfa8010aab000 0x1cb04000 0x1000 0xa5db000 pid.2748.dmp
0xfa8010b6b000 0x1ca44000 0x1000 0xa5dc000 pid.2748.dmp
0xfa8010bf9000 0x1cd53000 0x1000 0xa5dd000 pid.2748.dmp
0xffffffffd00000 0x100000 0x1000 0xa5de000 pid.2748.dmp
0xffffffffd01000 0x105000 0x1000 0xa5df000 pid.2748.dmp
0xffffffffd02000 0x101000 0x4000 0xa5e0000 pid.2748.dmp
0xffffffffd07000 0x3000 0x1000 0xa5e4000 pid.2748.dmp
0xffffffffd08000 0x1000 0x2000 0xa5e5000 pid.2748.dmp
0xffffffffd0b000 0x4000 0x3000 0xa5e7000 pid.2748.dmp
0xffffffffd0e000 0x0 0x1000 0xa5ea000 pid.2748.dmp
0xffffffffd0f000 0x7000 0x2000 0xa5eb000 pid.2748.dmp
```

然后从转储文件（pid.2748.dmp）中提取字符串（GET）

strings pid.2748.dmp | grep 'GET /'

```
0xffffffffd0b000 0x4000 0x3000 0xa5e7000 pid.2748.dmp
0xffffffffd0e000 0x0 0x1000 0xa5ea000 pid.2748.dmp
0xffffffffd0f000 0x7000 0x2000 0xa5eb000 pid.2748.dmp
0xffffffffd28000 0x40000 0x1f000 0xa5ed000 pid.2748.dmp
ubuntu@ip-172-31-24-202:~/Desktop/Start here/Artifacts$ strings pid.2748.dmp | grep 'GET /'
GET /rock/Plugins/cred64.dll HTTP/1.1
GET /rock/Plugins/clip64.dll HTTP/1.1
```

下载了两个 dll

5.识别这些附加组件的存储位置对于遏制和清理至关重要。恶意软件在其恶意活动中下载和使用的文件的完整路径是什么？

vol.py -f Windows\ 7\ x64-Snapshot4.vmem windows.filescan | grep -E "cred64\.dll|clip64\.dll"

```
ubuntu@ip-172-31-24-202:~/Desktop/Start here/Artifacts$ nano f.txt
ubuntu@ip-172-31-24-202:~/Desktop/Start here/Artifacts$ grep -E "cred64\.dll|clip64\.dll" f.txt
0x1d971940 \Users\0xSh3r10ck\AppData\Roaming\116711e5a2ab05\clip64.dll 216
ubuntu@ip-172-31-24-202:~/Desktop/Start here/Artifacts$
```

6.一旦获取到文件，恶意软件就会激活其附加组件。恶意软件会启动哪个子进程来执行这些文件？

vol.py -f Windows\ 7\ x64-Snapshot4.vmem windows.pstree

```
3028 2144 GoogleCrashHan 0xfa8001bfe6f0 5 81 0 False 2023-08-09 21:32:21.000000 N/A
2748 2524 lssass.exe 0xfa800300a750 7 254 1 True 2023-08-09 21:33:04.000000 N/A
* 3064 2748 rundll32.exe 0xfa8003042b30 1 64 1 True 2023-08-09 21:33:56.000000 N/A
ubuntu@ip-172-31-24-202:~/Desktop/Start here/Artifacts$
```

7.全面了解 Amadey 的持久化机制有助于有效缓解其威胁。除了已重点关注的位置之外，该恶意软件还可能在哪些地方确保其持续存在？

vol.py -f Windows\ 7\ x64-Snapshot4.vmem windows.filescan | grep "lssass.exe"

```
ubuntu@ip-172-31-24-202:~/Desktop/Start here/Artifacts$ vol.py -f Windows\ 7\ x64-Snapshot4.vmem windows.filescan | grep "lssass.exe"
0x517b290 100.0 \Users\0xSH3R-1\AppData\Local\Temp\925e7e99c5\lssass.exe 216
0x1dad11e0 \Windows\System32\Tasks\lssass.exe 216
0x1e994b20 \Users\0xSH3R-1\AppData\Local\Temp\925e7e99c5\lssass.exe 216
```